Computer Programming (NCSV101)
[Monsoon 2025-26]

Slides 1.2

Program Execution Process

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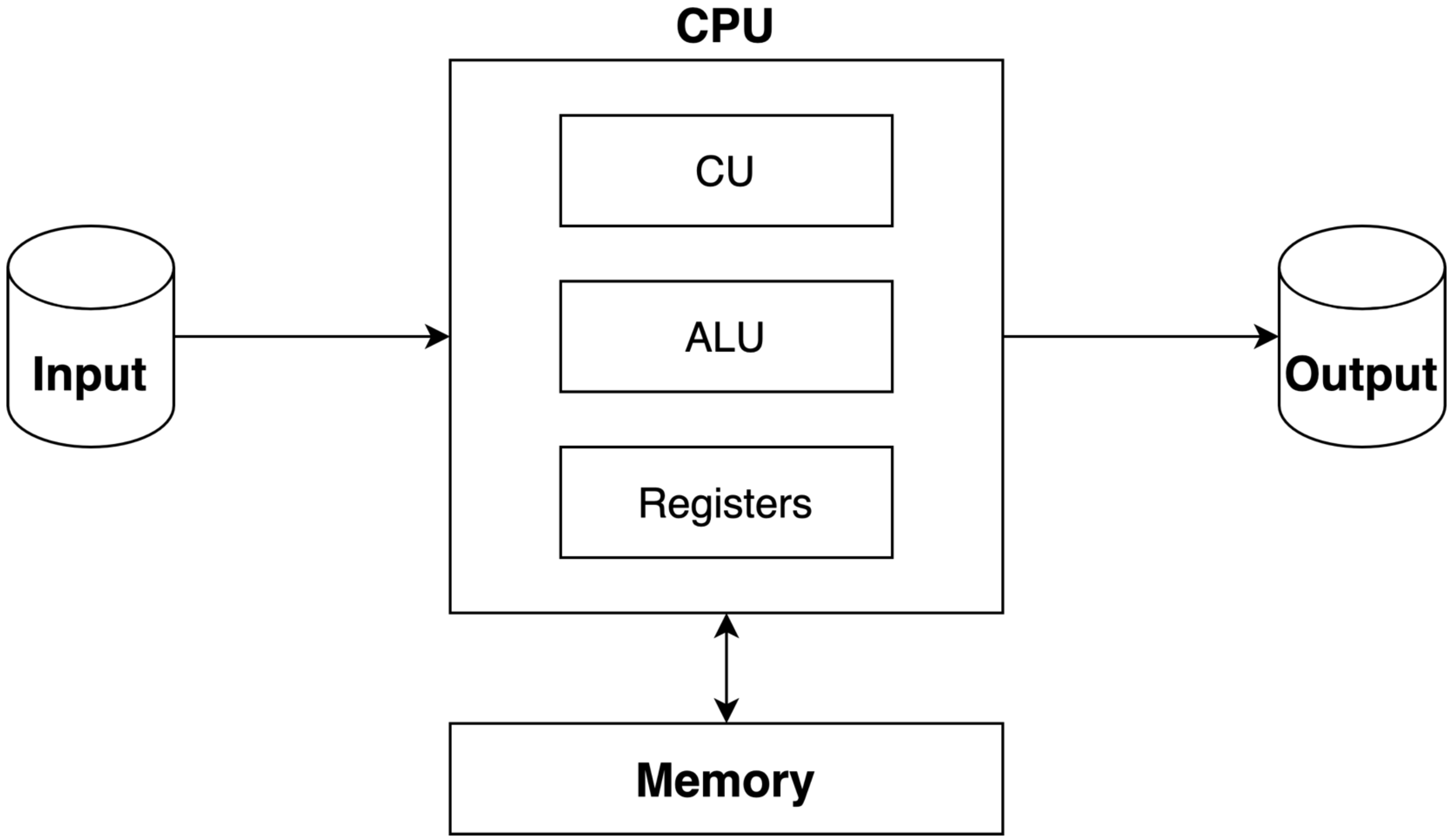
Revision

You now know the basic elements of a Computer

- CPU, Input Devices, Output Devices and a Memory Element

von Neumann Architecture



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But think about how tedious it will be to use these electronic components

- We are talking about 0s and 1s – that too, lots of them !!

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- We are talking about 0s and 1s – that too, lots of them !!

Modern Computers allow you to put an assistant for yourself which does this hard work
- We call this assistant the **Operating System** or **OS** – because it operates the “hardware” on our behalf
- Hardware is just a glorified term for all the underlying electronic components

The Operating System adds a layer over the hardware
- You can talk to the Operating System, asking it to get the computations done from the CPU
- It also manages your Memory element – we usually call it the **Main Memory**

Main Memory

Although not technically accurate, the colloquial term for Main Memory is “RAM”

The Main Memory is where the Operating System puts instructions for the CPU to execute

Any data, that is required to execute the instructions are also kept in the Main Memory

Main Memory
Capacity: 16 bytes

0000		00
0001		01
0010		02
0011		03
0100		04
0101		05
0110		06
0111		07
1000		08
1001		09
1010		10
1011		11
1100		12
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1110		14
1111		15

This is how the Main Memory looks like

Each Memory Location has an *address* (You can see the addresses here in both binary and decimal)

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This is how the Main Memory looks like

Each memory location can store 1 or more bytes – we call this the *word length* (as of now, assume the length to be 1 byte)

Each Memory Location has an *address* (You can see the addresses here in both binary and decimal)

Main Memory

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The Main Memory is where the Operating System puts instructions for the CPU to execute

Any data, that is required to execute the instructions are also kept in the Main Memory

The word length is dependent on the processor, and the number of wires in the *buses*

- Don’t worry much about this as of now, you will study about Word lengths in Computer Organisation
- For now, just assume that every address in the memory can store 1 byte !!

We can store both instructions, as well as data in the Main Memory

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
0100		04
0101		05
0110	0 0 0 0 1 0 0 0	06
0111		07
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Some part of the memory has instructions

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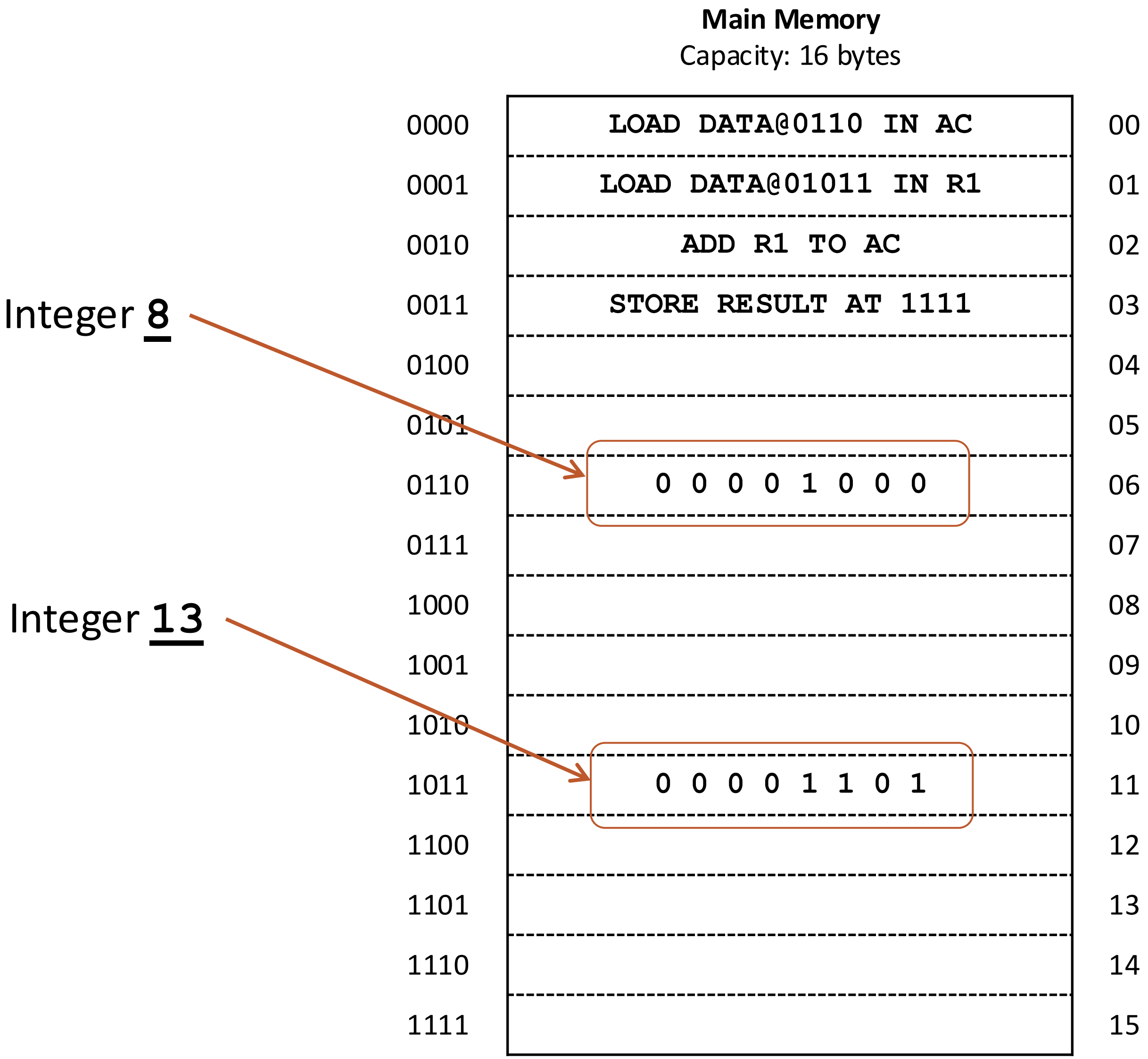
Some part of the memory has instructions

While other parts carry data



Of course, everything has to be stored in terms of 0s and 1s

Revisiting CPU – A sample program

Let us revisit our CPU again as well

Remember that we were going to instruct the Control Unit to perform computations?

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Remember that we were going to instruct the Control Unit to perform computations?

- Let us take the example of Addition
- We want the Control Unit to add two numbers – `8` and `13`

How can we arrange the relevant instructions and data in the Memory?

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
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CPU Registers

PC	X X X X X X X X
AC	X X X X X X X X
R1	X X X X X X X X

Remember the *Registers* we talked about?
Here are some examples !!

There are two special Registers named `PC` and `AC` here

… and there can be others, called `R1`, `R2`, `R3` and so on

Revisiting CPU – A sample program

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Remember that we were going to instruct the Control Unit to perform computations?

- Let us take the example of Addition
- We want the Control Unit to add two numbers – `8` and `13`

`AC` stands for Accumulator Register

- It is the primary register that ALU uses for its operations
- For any single operand operations, the Control Unit simply stores the operand in AC
- … and instructs the ALU to perform the operation
- For two operand operations, ALU can use one of the registers `R1`, `R2` etc.

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Remember that we were going to instruct the Control Unit to perform computations?

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- It is the primary register that ALU uses for its operations
- For any single operand operations, the Control Unit simply stores the operand in AC
- … and instructs the ALU to perform the operation
- For two operand operations, ALU can use one of the registers `R1`, `R2` etc.

`PC` stands for Program Counter

- It stores the address of *the next instruction to execute* – let us walk through an example to understand

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
0100		04
0101		05
0110	0 0 0 0 1 0 0 0	06
0111		07
1000		08
1001		09
1010		10
1011	0 0 0 0 1 1 0 1	11
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CPU Registers

PC	X X X X X X X X
AC	X X X X X X X X
R1	X X X X X X X X

Let us assume that the Control Unit wants to use the ALU for the addition operation

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
0100		04
0101		05
0110	0 0 0 0 1 0 0 0	06
0111		07
1000		08
1001		09
1010		10
1011	0 0 0 0 1 1 0 1	11
1100		12
1101		13
1110		14
1111		15

CPU Registers

PC	0 0 0 0 0 0 0 0
AC	X X X X X X X X
R1	X X X X X X X X

The first step is to set the value of `PC` to the beginning of the "addition program"

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
0100		04
0101		05
0110	0 0 0 0 1 0 0 0	06
0111		07
1000		08
1001		09
1010		10
1011	0 0 0 0 1 1 0 1	11
1100		12
1101		13
1110		14
1111		15

CPU Registers

PC	0 0 0 0 0 0 0 0
AC	X X X X X X X X
R1	X X X X X X X X

The first step is to set the value of `PC` to the beginning of the "addition program"

The instruction says – "load the data stored at address `0110` into the `AC` register"

Main Memory

Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
0100		04
0101		05
0110	0 0 0 0 1 0 0 0	06
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1111		15

CPU Registers

PC	0 0 0 0 0 0 0 1
AC	0 0 0 0 1 0 0 0
R1	X X X X X X X X

After this instruction gets executed, `AC` contains the data from the `0110` location

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
0100		04
0101		05
0110	0 0 0 0 1 0 0 0	06
0111		07
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CPU Registers

PC	0 0 0 0 0 0 0 1
AC	0 0 0 0 1 0 0 0
R1	X X X X X X X X

After this instruction gets executed, `AC` contains the data from the `0110` location

The `PC` register gets incremented by *1* after execution of every instruction – so it now has a value which points to the next instruction of the program

The next instruction also asks CU to load a value in a register – value at address `1011` into register `R1`

Main Memory
Capacity: 16 bytes

Address	Content	#
0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
0100		04
0101		05
0110	0 0 0 0 1 0 0 0	06
0111		07
1000		08
1001		09
1010		10
1011	0 0 0 0 1 1 0 1	11
1100		12
1101		13
1110		14
1111		15

CPU Registers

Register	Value
PC	0 0 0 0 0 0 1 0
AC	0 0 0 0 1 0 0 0
R1	0 0 0 0 1 1 0 1

The other operand for addition is now in another register – `R1`

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
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CPU Registers

PC	0 0 0 0 0 0 1 0
AC	0 0 0 0 1 0 0 0
R1	0 0 0 0 1 1 0 1

The other operand for addition is now in another register – `R1`

The PC register now points to the next instruction – at location `0010` in Memory

This time, the instruction asks CU to perform an arithmetic operation – addition of values in registers `R1` and `AC`

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
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0101		05
0110	0 0 0 0 1 0 0 0	06
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CPU Registers

PC	0 0 0 0 0 0 1 1
AC	0 0 0 1 0 1 0 1
R1	0 0 0 0 1 1 0 1

The other operand for addition is now in another register – `R1`

The PC register now points to the next instruction – at location `0010` in Memory

This time, the instruction asks CU to perform an arithmetic operation – addition of values in registers `R1` and `AC`

Now, the CU activates ALU, which adds the contents of the two registers, and overwrites it back in the AC register

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1111		15

CPU Registers

PC	0 0 0 0 0 0 1 1
AC	0 0 0 1 0 1 0 1
R1	0 0 0 0 1 1 0 1

Since,
$8 + 13 = (21)_{10}$
$= (10101)_2$

The other operand for addition is now in another register – R1

The PC register now points to the next instruction – at location 0010 in Memory

This time, the instruction asks CU to perform an arithmetic operation – addition of values in registers R1 and AC

Now, the CU activates ALU, which adds the contents of the two registers, and overwrites it back in the AC register

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CPU Registers

PC	0 0 0 0 0 0 1 1
AC	0 0 0 1 0 1 0 1
R1	0 0 0 0 1 1 0 1

The PC register now points to the next instruction – at location 0011 in Memory

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
0100		04
0101		05
0110	0 0 0 0 1 0 0 0	06
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1100		12
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CPU Registers

PC	0 0 0 0 0 0 1 1
AC	0 0 0 1 0 1 0 1
R1	0 0 0 0 1 1 0 1

The PC register now points to the next instruction – at location 0011 in Memory

This time, the instruction asks CU to store the "result" – which is basically the value of AC register, to the memory location 1111

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
0100		04
0101		05
0110	0 0 0 0 1 0 0 0	06
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1100		12
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1110		14
1111	0 0 0 1 0 1 0 1	15

CPU Registers

PC	0 0 0 0 0 0 1 1
AC	0 0 0 1 0 1 0 1
R1	0 0 0 0 1 1 0 1

The CU now transfers the content of `AC` to location `1110` of the Memory

Main Memory
Capacity: 16 bytes

0000	LOAD DATA@0110 IN AC	00
0001	LOAD DATA@01011 IN R1	01
0010	ADD R1 TO AC	02
0011	STORE RESULT AT 1111	03
0100		04
0101		05
0110	0 0 0 0 1 0 0 0	06
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CPU Registers

PC	X X X X X X X X
AC	X X X X X X X X
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The contents of the registers now *does not matter*, as the program has been executed successfully

Basically, at this point, the CU has essentially loaded the starting address of the next program to execute, and the cycle repeats

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Congratulations …
You are now a programmer !!

This "program" is a simplified version of an "assembly program" – done in an assembly language

The *Instructions* in an Assembly Program

We saw the instructions written as `LOAD`, `ADD` or `STORE`

However, in the memory, they too are just some “sequence of bits”

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The instruction `LOAD DATA@0110 IN AC` could be stored as `00011011`

- Here, `00` is the code for the *load operation*
- `0110` is the *location of the operand*
- …and `11` represents the AC register

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Other registers, like `R1` and `R2` may be represented by codes `01` and `10`

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- …and `11` represents the AC register

Other operations, like `STORE` and `ADD` may have codes like `01` and `11`

Other registers, like `R1` and `R2` may be represented by codes `01` and `10`

What differs, is how these bits are interpreted – they can be interpreted as instruction or data

The Input Device

We understand now that if a program is stored in the memory, the CU can execute it

But how do we store it there?

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But how do we store it there?

This is where *input devices* come into picture

A typical input device is keyboard

The Operating System “understands” how your “input device” works !!

The input devices are connected to the memory by buses

The Operating System contains “assembly programs to transfer data” from the device to memory

- That is all you need to know for now !! “Somehow” the OS knows when and how to do this

The Output Device

The sum that we did using the program needs to be "retrieved" too

An Output Device can fetch this data from the memory for us

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Again, just like the Input Device, the OS knows “how to manage” the device

A typical output device is your Laptop’s Screen or a Monitor

- Another popular output device is Printer

The Operating System has programs which can send data from the memory to the device

- Again… that is all you need to know at this stage !!

To summarise…

The Operating System is a *collection of programs* which can

- Manage Input Devices
- Manage Output Devices
- Manage Main Memory
- Co-ordinate with the Control Unit to get some computation done

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The Operating System is a *collection of programs* which can

- Manage Input Devices
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- Manage Main Memory
- Co-ordinate with the Control Unit to get some computation done

It provides an “easier” interface for us to get our jobs done

- Easier, because we can, in theory, directly manage the CPU as well !!
- But that will involve speaking in 0100100010101111 0110… Sorry I got a little carried away :P

You’ll have a whole subject dedicated to Operating Systems, so I leave it here for now

Homework !!

Read more about the type of operations that a typical Assembly Language may have

- Just reading this tutorial maybe more than enough for now: https://www.tutorialspoint.com/assembly_programming/assembly_introduction.htm